

Transfer Learning through Federated Reinforcement Learning

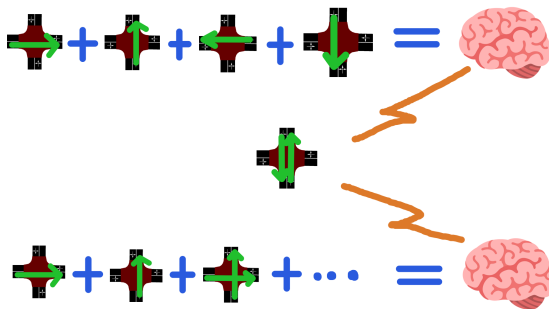
Machine Learning Paradigms Exam

Francesco Refolli



The Multi-Agent Reinforcement Learning Story So Far

Curriculum Learning



Monolithic Approach

Figure: Comparison of Approaches for Reaching Demand Coverage[2]

The Multi-Agent Reinforcement Learning Story So Far

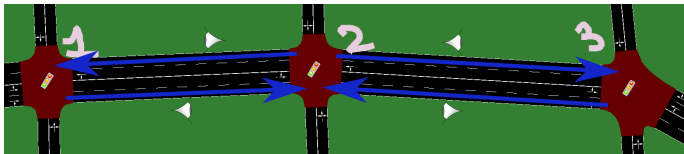


Figure: Agents sharing Observation [2]

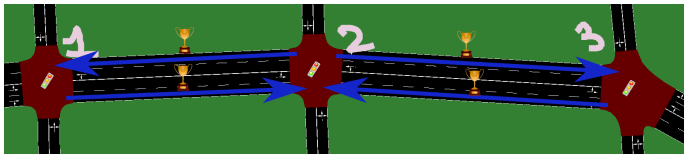
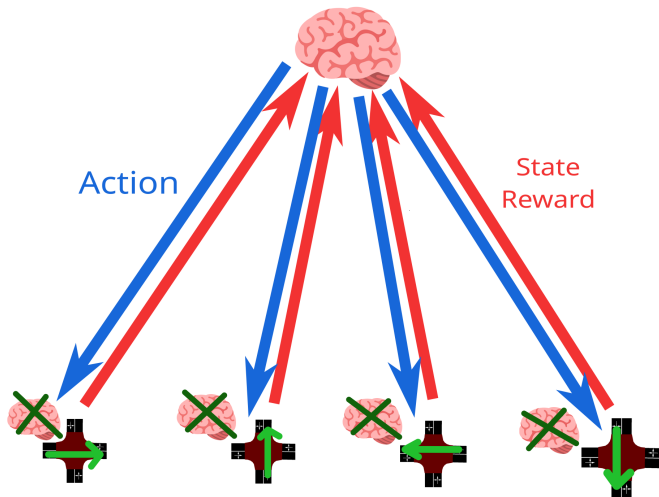
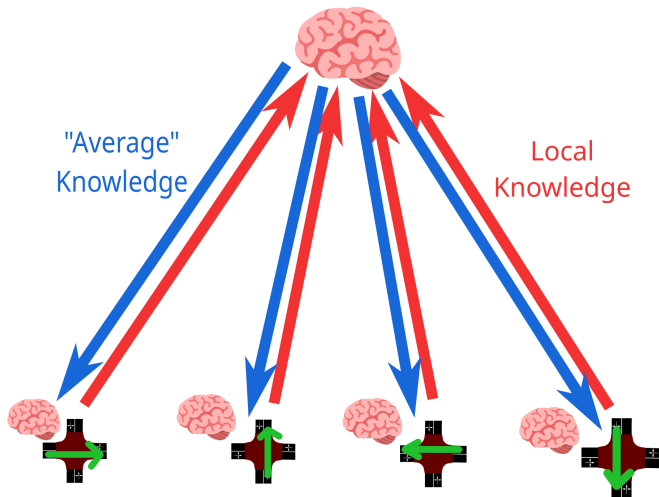


Figure: Agents sharing Rewards [2]

Centralized Reinforcement Learning

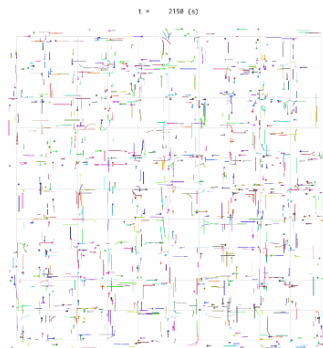
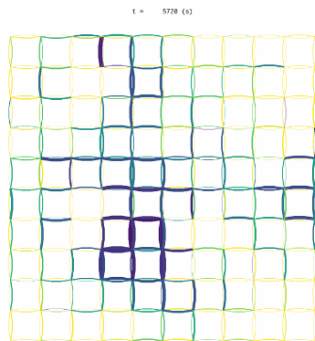


Federated Reinforcement Learning



The Simulator: UXsim

UXSim [3] is a Free and Open-Source *macroscopic* and *mesoscopic* network traffic flow simulator. It's tailored to simulate private-mobility simulations. It's "**Lightweight**", **Library-Based** and **Written in Python**.



The Agent Model

- The Agent can change Traffic Light Phase every 10 seconds
- Each one of the two *phases* gives *right of way* to one of the two flows
- The Agent can see the density of vehicles in each one of the roads (input/output)
- The Agent is rewarded based on vehicle queues' length, specifically on the amount vehicles en/dequeueing

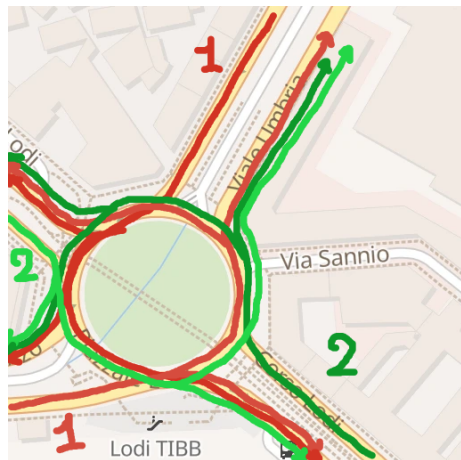


Figure: 4-Way Double-Lane Traffic Light Regulated Intersection

The Traffic Model

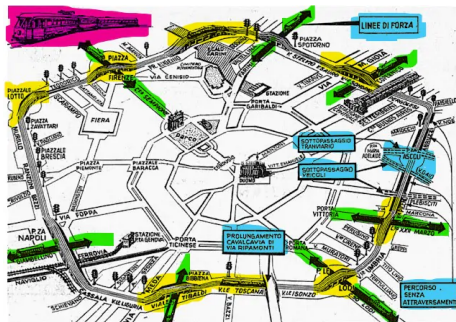


Figure: Ring-Road of a Metropolitan City [5]

- Two-Lanes per Road Link
- Road Length is 1000 m = 1 km
- Free Flow Speed is 20 m/s $\approx$ 72 km/h [4]
- Traffic Jam Density is set to $\rho = 0.2 \approx 40$ veh/2lanes/1km [4]
- No Platoons allowed, as in the Real-World (at least for now ...)
- No simulated Pedestrian Crossing, nor Public Transit (of any kind)

The Demand Model / Training Scenarios

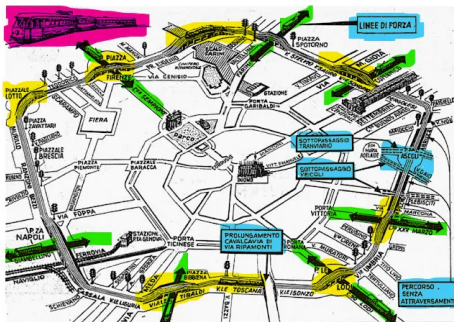


Figure: Ring-Road of a Metropolitan City [5]

- All Mid
(**Piazzale Fratelli Zavattari**)
- Peak East to West
(**Piazzale Loreto**)
- Peak West to East
(**Piazza Simone Bolivar**)
- Peak North to South
(**Piazza Firenze**)
- Peak South to North
(**Piazzale Lodi**)

The FedAvg Algorithm

Listing 1: FedAvg Algorithm

```
def Coordinator():
     $\theta$  = Policy.New()
     $\forall c \in C \mid \text{SendToClient}(c, \theta)$ 
    for  $e \in \{1, \dots, E\}$ :
         $\{(\theta_1, w_1) \dots\} = \{\text{GetFromClient}(c) \mid \forall c \in C\}$ 
         $\{\alpha_1 \dots\} = \{\frac{w_1}{\sum w_i} \dots\}$ 
         $\theta = \sum \alpha_i \theta_i$ 
         $\forall c \in C \mid \text{SendToClient}(c, \theta)$ 
    return  $\theta$ 

def Client(scenario):
    for  $e \in \{1, \dots, E\}$ :
         $\theta = \text{GetFromCoordinator}()$ 
         $\theta', w = \text{Simulation}(\theta, \text{scenario})$ 
         $\text{SendToCoordinator}(\theta')$ 
```

- Knowledge is weighted on the complexity of the scenario of a client
- ... which is equal to the demand that the client has to deal with
- Due to resource constraints, only 5 Peers were used, with no client sampling
- The duration of a simulation is $3600s = 1hr = 360$ steps.

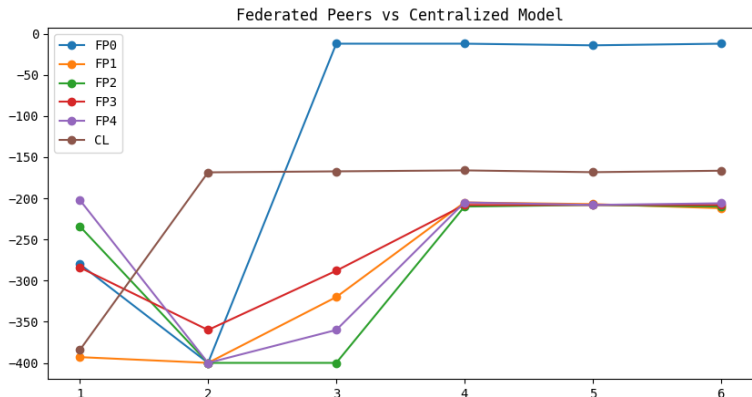


Figure: Total Reward during Training over Time (Episodes)

Results / Evaluation

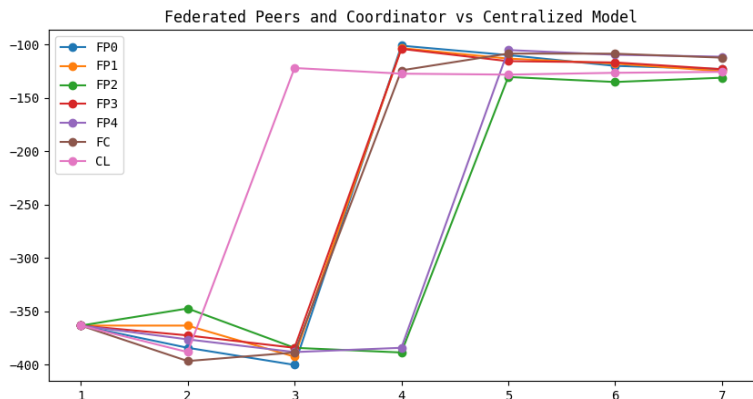


Figure: Average Total Reward during Evaluation over Time (Initial + Episodes)

Results / The Federated Coordinator is Catching Up!

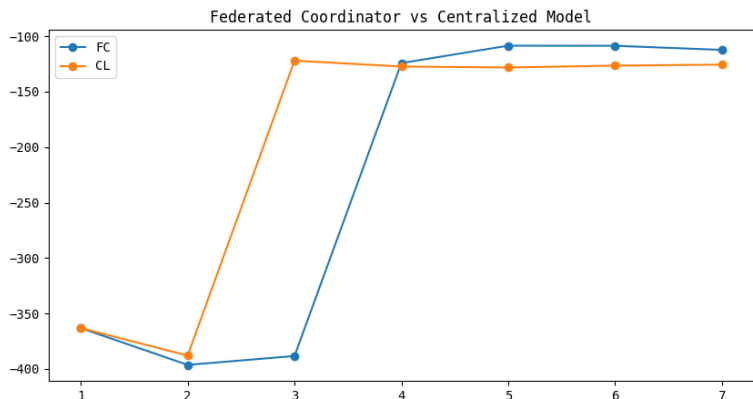


Figure: Average Total Reward during Evaluation over Time (Initial + Episodes)

Comparisons / Fixed-Cycle Agents

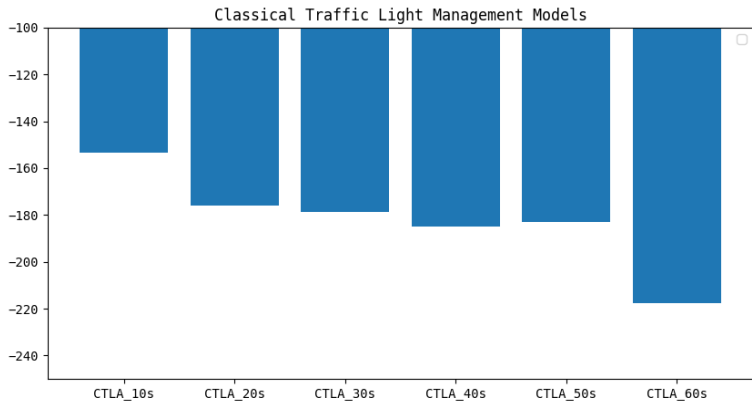


Figure: Average Total Reward during Evaluation of Fixed-Cycle Agents (*CTLA-Ts* means *T* seconds per phase)

Comparisons / Is Deep Q-Learning Worth the Effort?

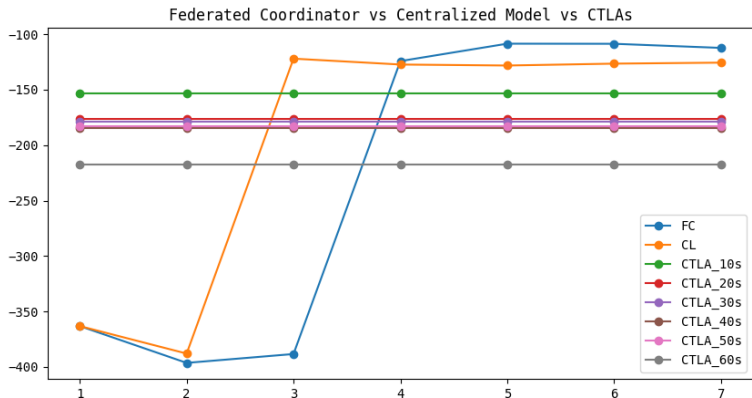


Figure: Comparison of Average Total Reward obtained by Deep Q-Learning Models and Fixed-Cycle Agents during Evaluation

Comparisons / Is Deep Q-Learning Worth the Effort?

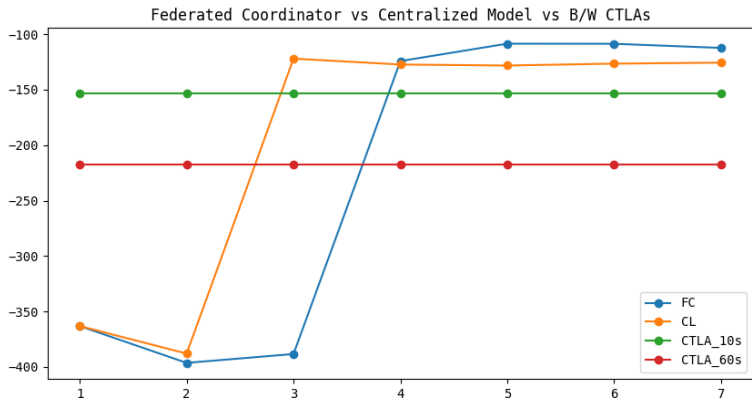


Figure: Comparison of Average Total Reward obtained by Deep Q-Learning Models and Fixed-Cycle Agents during Evaluation

Visualizing that Reward Difference

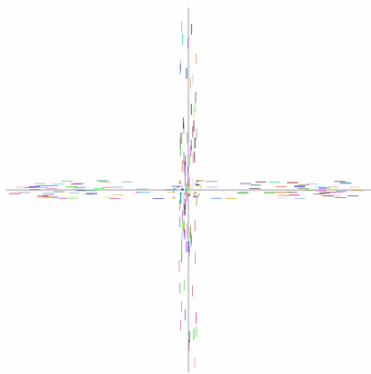


Figure: Microscopic visualization of Fixed-Agent (10s per phase)

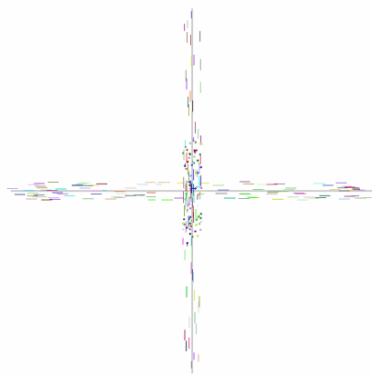


Figure: Microscopic visualization of Fixed-Agent (60s per phase)

Visualizing that Reward Difference

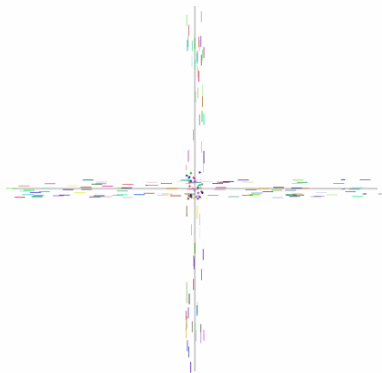


Figure: Microscopic visualization of Centralized Learning Agent

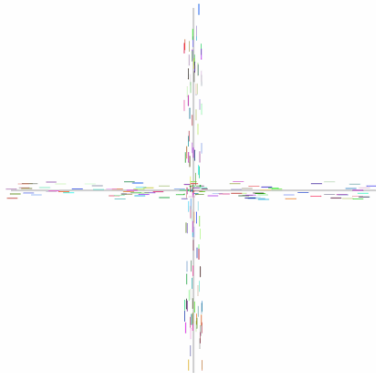


Figure: Microscopic visualization of Federated Learning Agent

Thank You

frefolli/**Machine-
Learning-Paradigms-...**



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Contributor



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Issues



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Stars



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Forks



- [1] TuttoCittà. *Map of Piazzale Lodi, Milan, Italy*. TuttoCittà. Accessed: April 20, 2026. Screenshot by author. 2026. URL: www.tuttocitta.it/mappa/milano/piazzale-lodi.
- [2] Refolli Francesco. “Unconventional Reinforcement Learning on Traffic Lights with Sumo”. In: September 2025. URL: <https://github.com/frefolli/master-presentation>.
- [3] Toru Seo. “UXsim: lightweight mesoscopic traffic flow simulator in pure Python”. In: *Journal of Open Source Software* 10.106 (2025), p. 7617. DOI: 10.21105/joss.07617. URL: <https://doi.org/10.21105/joss.07617>.
- [4] Takashi Nagatani. “The physics of traffic jams”. In: *Reports on progress in physics* 65.9 (2002), pp. 1331–1386.
- [5] Corriere della Sera. *Disegno del percorso della linea tranviaria circolare per Milano*. 1973.